MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TYPE 7792[1/]

The complete requirements for procuring the electron tubes described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: Diode, Hydrogen
Base: See figure 1
Super-Gumbo, 4 pin with bayonet, A4-106
Dimensions: See figure 1
Pin number: See figure 1

1	2	3	4	Cap
nc	h k	h	nc	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	ib	Ib	Ip	tk	Fault Electrode Current (0.1 sec max)	TA	Alt
Unit:	Vac	kv	a	Adc	Aac	sec	a	°C	feet
Rectifier:									
Maximum:	12.1	20 (Notes 1&2)	8	1.8 (Note 1)	---	---	60	+75	10,000
Minimum:	10.9	---	---	---	---	180	---	-55	---
Charging Diode:									
Maximum:	12.1	25	2.2	1.0	1.5	---	60	75	10,000
Minimum:	10.9	---	---	---	---	180	---	-55	---
TEST CONDITIONS:									
	11.5	---	---	---	---	180	---	---	---

GENERAL:

Qualification - Not required
Holding period - t=96 hours
Rough-handling end points - 1136 MIL-STD-1311

1/ Formerly 7791, 7792

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL	INSP. LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN.	MAX.	
	Quality conformance inspection, part 1							
1301	Heater current	---	0.65	II	If	6.75	8.25	Aac
1353	Operation of rectifier (1)	(Notes 4, 5, 7 and 8) Ef=10.9 Vac	0.65	II	---	---	---	---
1353	Operation of rectifier (2)	(Notes 4, 5, 7 and 8) Ef=12.1 Vac	0.65	II	---	---	---	---
3202	Starting voltage	(Notes 9 and 10)	0.65	II	Ez	---	100	V
3204	Voltage drop	(Notes 9 and 10)	0.65	II	etd	---	60	V
---	Operation of charging diode	(Notes 8 and 11)	0.65	II	---	---	---	---
	Quality conformance inspection, part 3							
---	Life test (1)	(Notes 4, 5 and 12)	---	---	t	1000	---	hr.
---	Life-test end points	Operation (2); (Note 8), Voltage drop	---	---	etd	---	80	V
	Periodic-check test							
1353	Operation of rectifier (3)	Operation of rectifier (2); (Note 3)	---	---	---	---	---	---
---	Life test (2)	(Notes 6 and 11)	---	---	t	200	---	hr.
---	Life-test end points	Starting voltage	---	---	Ez	200	---	V

NOTES:

1. Maximum current and voltage ratings do not apply simultaneously. *See figure 2 for current versus* voltage ratings.
2. Whenever possible when applying anode voltage it is desirable to gradually increase the voltage to the desired operating value. Anode voltage in excess of 15 kv should never be abruptly applied.
3. Ambient temperature 75°C. The tube shall operate for 1 hour without flashing or arc back.
4. Circuit constants: epx=22 kv; ib=8 a; Ib=2 Adc. Minimum fault current during arc back shall be 60 amperes.
5. This test shall be performed with an air flow of 50 cfm directed down on top of radiator.
6. Life test shall be performed on a minimum of two tubes.
7. At t=180, raise plate voltage from zero to specified value in approximately 15 seconds.
8. The tube shall go into conduction smoothly and operate for 10 minutes without arc back, sparking, flicker, or abnormal glow discharge. Three arcbacks are permitted prior to an uninterrupted 10 minute operation run.
9. *Low voltage (110 - 660) 60 cycle,* half wave rectifier circuit where I_o=2 Adc; ib=8 a.
10. This value shall be determined by means of a calibrated oscilloscope connected between anode and cathode after tube has been conducting for 120 seconds.

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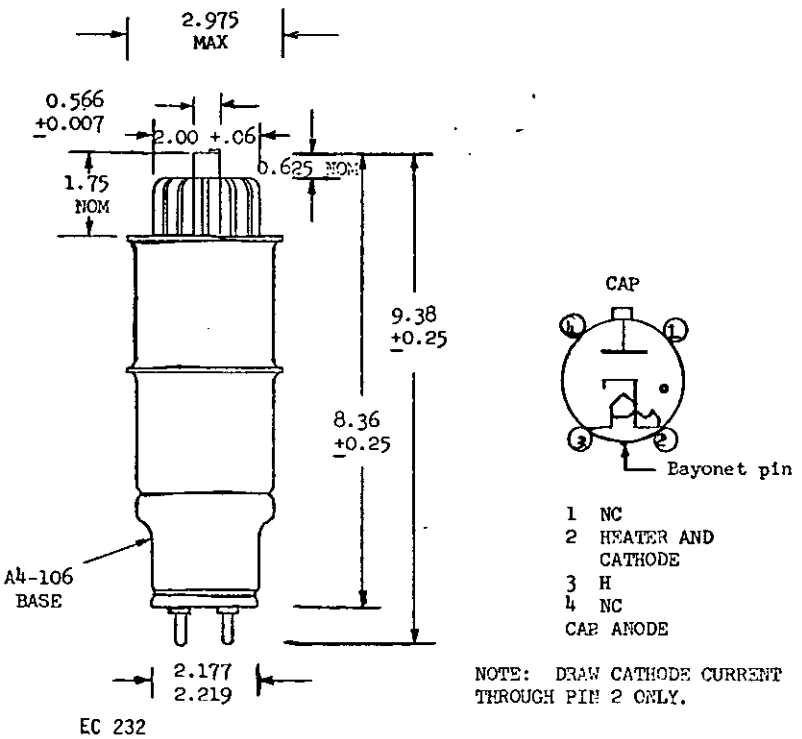
11. The tube shall be operated as a charging diode in a thyratron modulator. The circuit constants shall be chosen to give the following diode operating conditions: network forward voltage 25kv, ib=2.2a±10% and average current (Ib)=1.0A minimum.
12. Life test shall be performed on a minimum of one tube per month for a maximum production lot of 250 tubes.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy-EC

(Project 5960-N808)

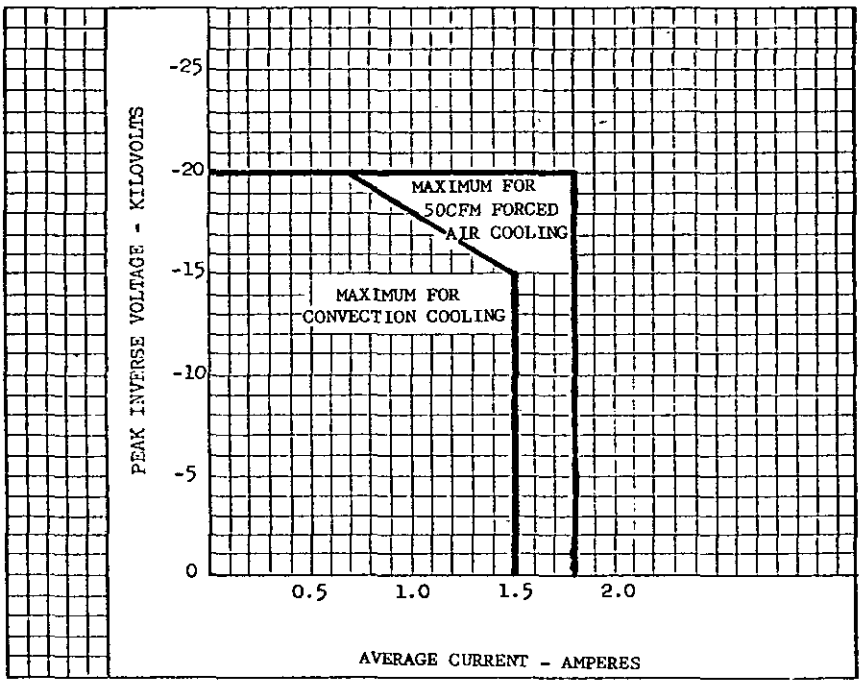
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ALL DIMENSIONS IN INCHES.

Figure 1. Outline drawing of electron tube type 7792.

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EC 233

Figure 2. Current versus voltage ratings.

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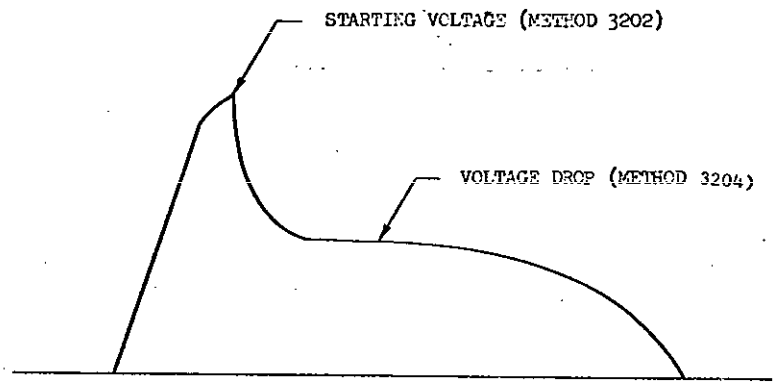


Figure 3. Anode-cathode voltage waveform.

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